Supplementary table 1: Detail of barriers and enablers to implementation of RGCS programmes

Theme	Sub theme	Barriers	Enablers
The use of reproductive carrier screening and the impact it may have	a) Achieving equitable service provision	Cost of testing Lack of consistency in panels public/private sector	Link with other health interventions Communication with policy makers and other stakeholders
	b) Potential impact (including the offer) on the patient	Possibility of raising women's anxiety Feeling coerced Lack of general public interest in RCS Misperceptions of the 'perfect child'	Informed consent – active participation and understanding implications Decision aids/client education Raise knowledge of general population
Practitioner beliefs and expectations about delivery of RCS	a) Practitioner attitudes to and beliefs about RCS	Lack of collective sense of urgency Positive attitude to RCS not followed up by offering RCS in practice Practitioners not interested Variability about who should be tested Concern about offering testing as routine would influence clients' decision to take up offer Concern about false positives/ medicalising pregnancy and perceptions of eugenics Unsure about adult onset conditions	Client request for RCS Experience
	b) Practitioner perceptions of their ability to deliver RCS	GPs – perception of lack of ability to discuss positive results and possibility of termination when women is first pregnant Lack of confidence in offering prenatal genetic advice, what diseases to test for, interpreting the results and explaining results Confusing criteria to guide HCPs makes RCS appear too hard	Obstetricians and gynaecologists were comfortable with offering RCS Training Knowing when to refer on
	c) Practitioner knowledge and support required to deliver RCS	Limited training in genetics leads to lack of underlying knowledge Lack of support for practitioners	Call for training Dissemination of research findings Raise awareness of screening guidelines Increase genetic support for HCPs
	d) Practitioner expectations and external views influencing their decision making	Offers made due to liability and professional obligation i.e. not guided by client need/preference Continual renegotiation of the place of RCS Conflicting advice from different professional bodies	Role of the professional bodies to provide consistent guidance Evidence based implementation including relevance and acceptance by society
Resources available for practitioners to use for reproductive carrier screening	a) Provision of counselling including genetic counsellors and other professionals	Access to counselling as a limited resource Concern non-specialist staff would find counselling challenging and underestimate the complexity of counselling	Ensure couples are well informed before a consultation Experience and time for non-specialist staff to develop time efficient counselling skills
	b) Variation in potential models of service provision	GPs-RCS does not fit with GP targets Use of repeat visits to ensure testing does not become routine	Develop GP specialists with screening skills Identify other professions to offer screening e.g. midwives Offer testing outside health care settings Preference to offer screening pre pregnancy, at preconception carrier screening clinics
	c) Non clinical resource barriers	Time Cost to the health system Lack of focus on public health Identifying appropriate measures of effectiveness	Upskilling HCPs (to reduce time) Additional resources Leadership Development of an implementation plan with implementation strategies